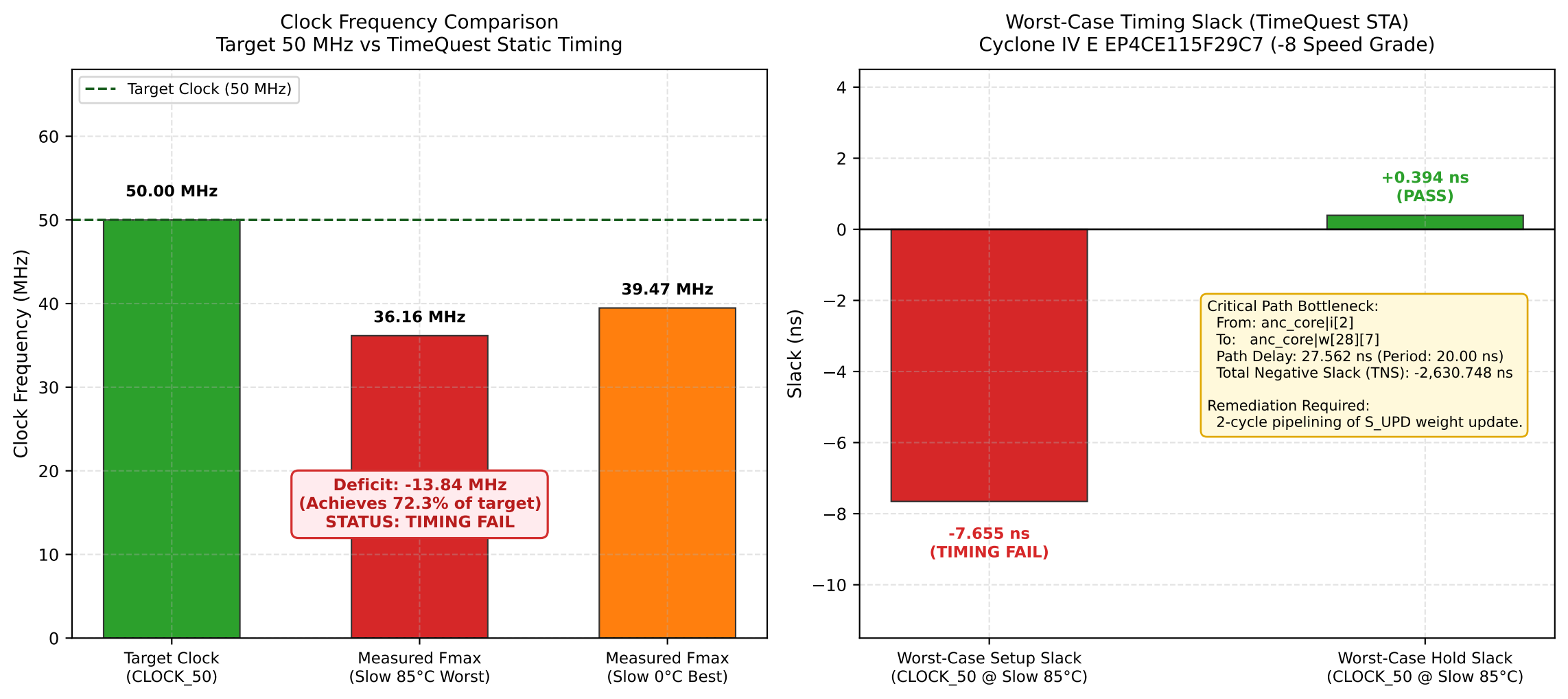
Clock Frequency Comparison
Target 50 MHz vs TimeQuest Static Timing
Target Clock (50 MHz)
Clock Frequency (MHz)
0
10
20
30
40
50
60
50.00 MHz
36.16 MHz
39.47 MHz
Deficit: -13.84 MHz
(Achieves 72.3% of target)
STATUS: TIMING FAIL
Target Clock
(CLOCK_50)
Measured Fmax
(Slow 85°C Worst)
Measured Fmax
(Slow 0°C Best)
Worst-Case Timing Slack (TimeQuest STA)
Cyclone IV E EP4CE115F29C7 (-8 Speed Grade)
Slack (ns)
4
2
0
−2
−4
−6
−8
−10
+0.394 ns
(PASS)
-7.655 ns
(TIMING FAIL)
Critical Path Bottleneck:
From: anc_core|i[2]
To: anc_core|w[28][7]
Path Delay: 27.562 ns (Period: 20.00 ns)
Total Negative Slack (TNS): -2,630.748 ns
Remediation Required:
2-cycle pipelining of S_UPD weight update.
Worst-Case Setup Slack
(CLOCK_50 @ Slow 85°C)
Worst-Case Hold Slack
(CLOCK_50 @ Slow 85°C)